

MANGNAI, China

WMO: 518860

Lat: 38.250N

Lon: 90.850E

Elev: 9662

StdP: 10.24

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.5	(c) 0.1	(d) -29.5	(e) 1.5	(f) 18.5	(g) -24.9	(h) 2.0	(i) 15.5	(j) 28.2	(k) 26.0	(l) 20.2	(m) 26.8	(n) 1.6	(o) 140	(p) 0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.0	(c) 79.7	(d) 49.6	(e) 76.8	(f) 48.6	(g) 74.2	(h) 47.5	(i) 53.7	(j) 69.5	(k) 51.4	(l) 68.8	(m) 49.8	(n) 68.1	(o) 7.8	(p) 70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 47.5	(b) 70.1	(c) 59.5	(d) 44.5	(e) 62.6	(f) 57.8	(g) 41.9	(h) 56.5	(i) 57.2	(j) 26.6	(k) 69.6	(l) 24.9	(m) 68.8	(n) 23.8	(o) 67.9	(p) 61.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.2	14.6	-7.2	85.1	2.7	2.8	-9.1	87.2	-10.7	88.8	-12.2	90.4	-14.2	92.5
			-8.8	55.5	2.4	3.0	-10.6	57.7	-12.0	59.4	-13.4	61.1	-15.1	63.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.4	14.2	22.5	32.6	43.1	51.4	58.7	63.4	62.3	53.4	40.1	26.7	15.8
	DBStd	17.98	5.80	6.45	6.42	6.48	5.19	4.49	4.79	4.31	4.59	5.51	6.20	5.22
	HDD50	4775	1111	771	539	222	46	2	0	0	16	309	700	1059
	HDD65	9030	1576	1191	1004	658	421	191	88	107	348	772	1150	1524
	CDD50	1284	0	0	0	14	90	264	416	380	118	2	0	0
	CDD65	65	0	0	0	0	0	3	39	22	1	0	0	0
	CDH74	630	0	0	0	1	5	49	346	223	6	0	0	0
	CDH80	80	0	0	0	0	0	2	51	27	0	0	0	0
Wind	WSAvg	5.4	3.4	4.6	6.3	7.1	6.8	6.6	6.3	6.1	5.5	4.6	3.9	3.3
Precipitation	PrecAvg	1.9	0.0	0.0	0.0	0.1	0.2	0.6	0.6	0.2	0.1	0.0	0.0	0.0
	PrecMax	4.2	0.2	0.1	0.2	0.6	1.4	2.1	1.2	1.0	0.5	0.2	0.1	0.1
	PrecMin	0.7	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.8	0.0	0.0	0.0	0.1	0.3	0.5	0.3	0.2	0.1	0.0	0.0	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.4	46.7	59.6	68.9	73.5	78.2	84.9	83.2	74.6	63.4	49.5	36.5
		MCWB	23.7	29.7	36.1	40.7	43.3	47.3	53.5	51.7	47.2	39.3	31.8	24.6
	2%	DB	31.7	41.8	53.8	64.5	70.0	75.2	80.7	79.1	70.8	59.3	45.3	32.1
		MCWB	21.0	26.9	32.9	38.9	42.0	46.4	50.3	49.2	45.2	37.2	29.4	21.9
	5%	DB	28.2	38.2	50.1	60.7	67.0	72.7	77.9	76.2	67.6	55.9	42.1	28.8
		MCWB	18.9	24.8	31.1	37.3	40.9	45.6	49.6	48.3	43.3	35.8	27.6	19.6
	10%	DB	25.0	34.8	46.4	57.5	64.0	70.0	74.9	73.6	64.9	52.6	38.9	26.0
		MCWB	16.8	23.0	29.4	35.6	39.9	44.6	48.5	47.5	42.1	34.2	25.9	17.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.1	29.9	36.5	41.3	45.2	51.2	57.6	56.2	51.5	40.5	32.1	25.2
		MCDB	35.4	45.7	56.9	66.1	67.1	66.9	75.4	73.8	65.3	56.4	48.4	35.5
	2%	WB	21.4	27.2	33.7	39.5	43.1	49.2	54.7	53.0	47.6	38.4	29.7	22.1
		MCDB	31.2	40.9	52.3	62.3	65.2	64.9	72.2	68.1	64.9	56.2	44.2	31.5
	5%	WB	19.0	25.1	31.5	37.8	41.9	47.7	52.2	51.1	45.6	36.5	27.9	19.8
		MCDB	27.8	37.7	49.2	58.8	63.2	64.5	70.0	68.4	62.6	54.5	41.6	28.5
	10%	WB	16.9	23.1	29.6	36.2	40.8	46.6	50.5	49.5	43.9	34.8	26.1	17.9
		MCDB	24.7	34.5	45.8	56.0	61.2	64.6	68.8	68.3	60.8	51.7	38.6	25.7
Mean Daily Temperature Range	5% DB	MDBR	21.8	23.3	24.8	24.7	22.2	20.0	20.0	21.3	22.8	24.6	23.5	21.2
		MCDBR	23.9	26.4	29.5	28.8	26.5	24.4	24.6	25.3	26.4	27.3	26.0	22.6
		MCWBR	16.6	16.4	16.2	14.0	11.6	9.3	8.6	9.3	11.3	13.6	15.6	15.6
	5% WB	MCDBR	23.3	25.7	28.0	26.5	23.0	18.7	20.1	19.7	20.7	25.5	25.4	22.0
		MCWBR	16.2	16.0	15.5	12.8	10.2	7.4	7.4	7.8	9.0	12.7	15.3	15.3
Clear-Sky Solar Irradiance	taub		0.228	0.269	0.324	0.389	0.413	0.427	0.415	0.396	0.353	0.288	0.246	0.223
	taud		2.444	2.315	2.112	1.924	1.925	1.956	2.053	2.095	2.182	2.313	2.413	2.461
	Ebn at Noon		312	306	295	281	275	271	273	276	282	294	300	306
	Edh at Noon		25	33	46	59	60	58	52	49	42	33	26	23
All-Sky Solar Radiation	RadAvg		948	1249	1685	2021	2214	2164	2083	2005	1766	1467	1082	870
	RadStd		52	52	56	101	97	137	138	80	80	53	43	35

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	+4.14	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page